



ITM SKILLS
UNIVERSITY

Computer Networks

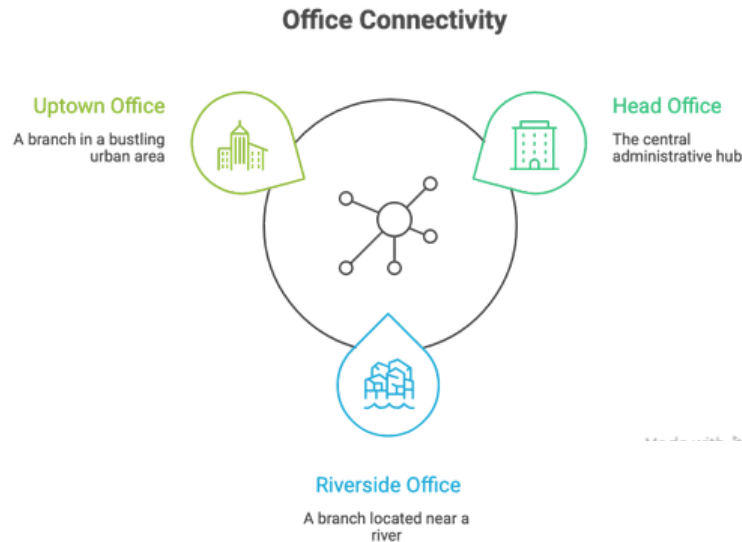
Harbor lane Legal Group

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Project Overview

Harbor Lane Legal Group requires a scalable network connecting its:

- Head Office
- Riverside Office
- Uptown Office



Head Office Requirements

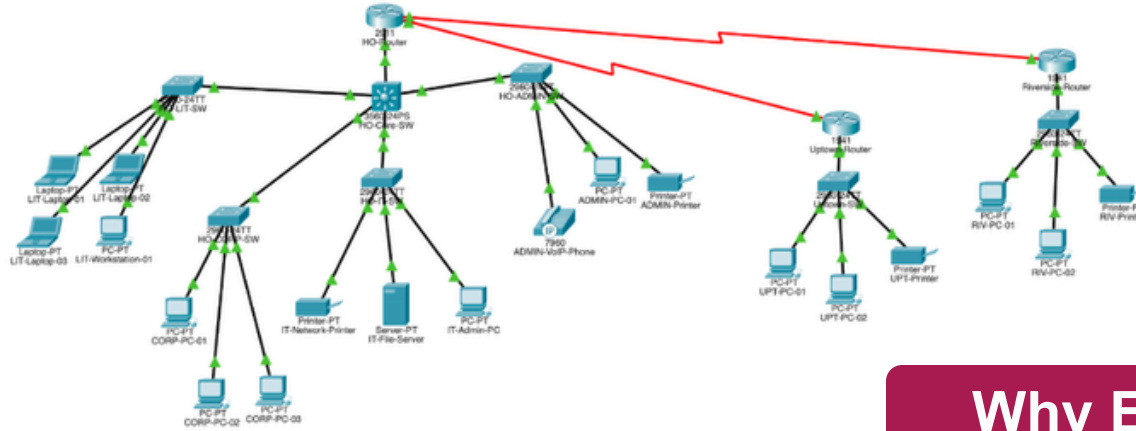
	Hosts
Litigation	45
Corporate Law	33
IT & Records	9
Reception/Admin	4
Voice Network	VoIP

Design Goals

Scalability • Segmentation • Efficient IP Addressing • Reliable Connectivity



NETWORK TOPOLOGY



Head Office: Extended Star

Satellite Offices: Star

WAN: Point-to-Point

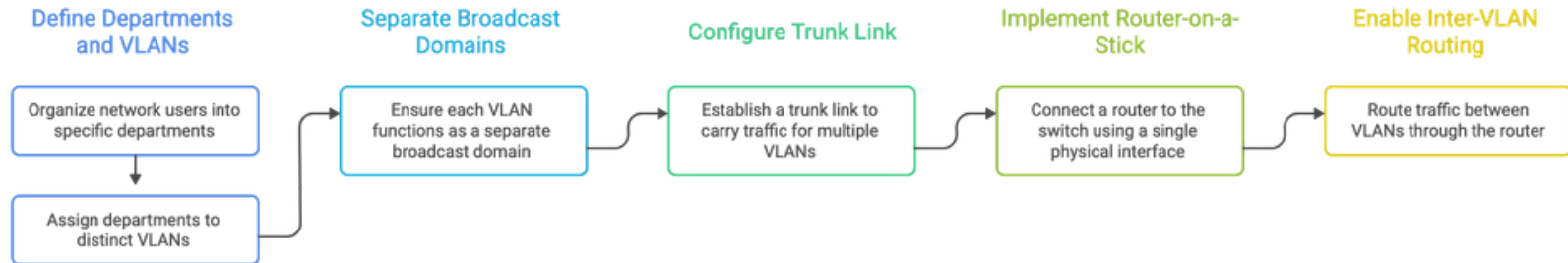
Why Extended Star?
Centralized Management
Easy Troubleshooting
Fault Isolation
Scalable

VLAN + ROUTER-ON-A-STICK

VLAN Department Mapping



Implement Inter-VLAN Routing



VLANs provide logical segmentation, while Router-on-a-Stick enables communication between VLANs.

VLSM - Variable length Subnet Mask

Network Segments and CIDR

Segment	Network	CIDR
 Litigation	192.168.40.0	/26
 Corporate	192.168.40.64	/26
 IT	192.168.40.128	/28
 Admin	192.168.40.144	/28
 Voice	192.168.40.160	/28
 Riverside WAN	192.168.43.0	/30
 Uptown WAN	192.168.43.4	/30

VLSM allocates different subnet sizes according to actual host requirements, reducing unnecessary IP address waste.

45 hosts → /26

33 hosts → /26

9 hosts → /28

4 hosts → /28*

WAN → /30

* /29 is technically sufficient for Admin; /28 was implemented for additional headroom.



PROTOCOLS + RESULTS

Network Protocol Functions

Protocol	Function
 DHCP	IP Configuration
 ARP	IP → MAC
 DNS	Name → IP
 ICMP	Connectivity Test

```
C:\>ping 192.168.40.130
```

```
Pinging 192.168.40.130 with 32 bytes of data:
```

```
Reply from 192.168.40.130: bytes=32 time<1ms TTL=127
Reply from 192.168.40.130: bytes=32 time<1ms TTL=127
Reply from 192.168.40.130: bytes=32 time<1ms TTL=127
Reply from 192.168.40.130: bytes=32 time=1ms TTL=127
```

```
Ping statistics for 192.168.40.130:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

**A scalable, segmented and
successfully tested network
connecting all three locations.**



Thank You